

Multivariable Calculus

MATH 1920 Lecture Notes

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1 Planes and Surfaces

1.1 Determining a Plane

A plane $\mathcal{P}$ in $\mathbb{R}^3$ can be uniquely determined by any of the following geometric data:

- A point Q on the plane and a normal vector $\vec{n} \perp \mathcal{P}$.
- Three non-collinear points.
- A general linear equation $ax + by + cz + d = 0$.
- Two intersecting lines (or two non-parallel direction vectors).
- Two parallel distinct lines.
- A line $\mathcal{L}$ and a point $Q \notin \mathcal{L}$.

1.2 Mathematical Formulations of a Plane

Case 1: Plane passing through the origin $O(0, 0, 0)$

A vector $\vec{v} = \langle x, y, z \rangle$ lies in $\mathcal{P} \iff \vec{v} \perp \vec{n}$, where $\vec{n} = \langle a, b, c \rangle$:

$$\vec{v} \in \mathcal{P} \iff \vec{v} \cdot \vec{n} = 0 \iff ax + by + cz = 0$$

In set notation:

$$\mathcal{P} = \{ \langle x, y, z \rangle \mid ax + by + cz = 0 \}$$

Case 2: Plane passing through a point

$Q(x_0, y_0, z_0)$

Let Q be a known point on $\mathcal{P}$ with position vector $\overrightarrow{OQ} = \langle x_0, y_0, z_0 \rangle$. Any point $W(x, y, z)$ on the plane satisfies:

$$(\overrightarrow{OW} - \overrightarrow{OQ}) \perp \vec{n} \iff (\overrightarrow{OW} - \overrightarrow{OQ}) \cdot \vec{n} = 0$$

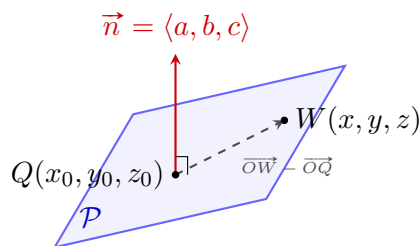
Expanding this dot product yields the point-normal form:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

or equivalently:

$$ax + by + cz = ax_0 + by_0 + cz_0$$

We can always scale the coefficients a, b, c, d in $ax + by + cz = d$ by $\frac{1}{\|\vec{n}\|}$ to obtain a unit normal vector $\hat{n} = \frac{\langle a, b, c \rangle}{\sqrt{a^2 + b^2 + c^2}}$.



1.3 Properties of Normal Vectors

For any set of vectors $\{\vec{u}, \vec{v}, \vec{w}\}$ where $\vec{v} \nparallel \vec{w}$, the following statements are equivalent:

- $\vec{u} \perp \vec{v}$ and $\vec{u} \perp \vec{w}$.
- $\vec{u} \parallel (\vec{v} \times \vec{w})$.
- $\vec{u} \perp \mathcal{P}$, where $\mathcal{P} = \text{Span}\{\vec{v}, \vec{w}\}$.

Problem 1.1: Finding Normal Vectors

Find a normal vector $\vec{n}$ to the plane $\mathcal{P}$ given by the equation $(x - 1) - (y - 1) = 0$.

Solution 1.1

The equation simplifies to $x - y = 0$. We can extract or compute the normal vector in two ways:

Method 1 (Direct Inspection): From the general form $ax + by + cz = d$, the coefficients of x, y, z directly give the normal vector:

$$\vec{n} = \langle 1, -1, 0 \rangle$$

Method 2 (Cross Product of Parallel Plane Vectors): Two vectors parallel to the plane $\mathcal{P}$ are $\vec{v}_1 = \langle 0, 0, 1 \rangle$ and $\vec{v}_2 = \langle 1, 1, 0 \rangle$. Taking their cross product gives:

$$\vec{n} = \vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \vec{i}(0 - 1) - \vec{j}(0 - 1) + \vec{k}(0 - 0) = \langle -1, 1, 0 \rangle$$

Note that $\langle -1, 1, 0 \rangle = -\langle 1, -1, 0 \rangle$, which represents the same normal line direction.

Problem 1.2: Finding a Plane From Three Points

Find the equation of the plane containing the points $P(1, 0, 1)$, $Q(0, 2, 0)$, and $R(2, 1, 0)$.

Solution 1.2

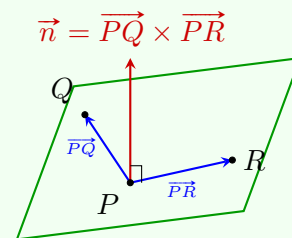
Step 1: Construct two displacement vectors in the plane

$$\overrightarrow{PQ} = \langle 0 - 1, 2 - 0, 0 - 1 \rangle = \langle -1, 2, -1 \rangle$$

$$\overrightarrow{PR} = \langle 2 - 1, 1 - 0, 0 - 1 \rangle = \langle 1, 1, -1 \rangle$$

Step 2: Compute the normal vector via cross product

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 2 & -1 \\ 1 & 1 & -1 \end{vmatrix} = \langle -1, -2, -3 \rangle$$



Step 3: Write the plane equation using point $P(1, 0, 1)$

$$-1(x - 1) - 2(y - 0) - 3(z - 1) = 0$$

Simplifying yields:

$$x + 2y + 3z = 4$$

What data determines a sphere?

- point in $\mathbb{R}^3$ and a number in $\mathbb{R}^{>0}$ (the radius)
- 2 points (the endpoints of a diameter)
- 4 points (noncoplanar)
- a plane tangent to the sphere and 3 points on the sphere

Definition 1.1: Traces

A trace is an intersection of an object with plane $x = k$, $y = k$, or $z = k$.

Example 1.1. Trace of a sphere is a circle.

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

Taking a trace of the sphere with the plane $z = k$, there are 3 cases:

- $|k - c| = R$: the trace is a single point at (a, b, k) .
- $|k - c| < R$: the trace is a circle of radius $\sqrt{R^2 - (k - c)^2}$ centered at (a, b, k) .
- $|k - c| > R$: the trace is empty (no intersection).

$\mathcal{P} = \{(x, y, z) | -2x + 6y - 5z = 4\}$, trace with the plane $x = -1$ is

$$-2(-1) + 6y - 5z = 4$$

$$2 + 6y - 5z = 4$$

$$6y - 5z = 2$$

$$z = \frac{6y}{5} - \frac{2}{5}$$

A quadric surface is given by

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fzx + ax + by + cz + d = 0$$

In practice, we often consider the quadric surface as a simpler form by assuming $D = E = F = 0$. This is what we often refer to as the “standard position” of a quadric surface.

Example 1.2. $x^2 + z^2 = 1$ is a cylinder centered on the y -axis with radius 1.

The $z = 0$ trace: $x = \pm 1$

The $x = 0$ trace: $z = \pm 1$ This implies the shape of the surface.

Example 1.3. $x^2 + 4y^2 - z^2 = -1$.

The $x = 0$ trace: $4y^2 - z^2 = -1$, $z = \pm\sqrt{4y^2 + 1} \approx \pm 2y$ for large enough $|y|$.

The $y = 0$ trace: $x^2 - z^2 = -1$, $z = \pm\sqrt{x^2 + 1} \approx \pm x$ for large enough $|x|$.

The $z = k$ trace: $x^2 + 4y^2 - k^2 = -1$.

- $|k| = 1$: the trace is a single point at the origin.
- $|k| < 1$: the trace is empty (no intersection).
- $|k| > 1$: the trace is an ellipse of the form $x^2 + 4y^2 = k^2 - 1$.

but now, how do we know that the ellipse would exactly fall on the surface?

$x^2 + y^2 + z^2 = 4$ intersection with line through $(1, 0, 1), (-1, 0, 1)$

line: $\ell(t) : (t, 0, 1)$ substitute $\ell(t)$ into the sphere equation:

$$(t)^2 + (0)^2 + (1)^2 = 4t^2 + 1 = 4t^2 = 3t = \pm\sqrt{3}$$

So the intersection points are $\ell(\sqrt{3}) = (\sqrt{3}, 0, 1)$ and $\ell(-\sqrt{3}) = (-\sqrt{3}, 0, 1)$.